

Aditya Tripathi

Burnaby, BC | adityarajtripathi3@gmail.com | [linkedin.com/in/aditya-tripathi](https://www.linkedin.com/in/aditya-tripathi) | github.com/adityaT39

Computer Science graduate with hands-on experience building and shipping full-stack, AI-powered web applications (Next.js/TypeScript, React, Python/FastAPI, Claude API). Seeking Junior Full Stack Software Engineer roles.

Education

University of British Columbia
Bachelor of Computer Science

2020 – 2024
Kelowna, BC

Relevant Coursework: Data Structures & Algorithms, Object-Oriented Design, Database Systems, Software Engineering, Human-Computer Interaction

Experience

Onpoint Consulting NZ

Jan 2026 – Jul 2026

Full Stack Software Developer (Contract)

Remote

- Designed and built two production websites end-to-end as **sole developer** on **Next.js, TypeScript, and Supabase (PostgreSQL)**: the corporate site and OnPoint Talent, a full-stack recruitment platform with **Stripe** billing and **Google OAuth**.
- Built a suite of **AI-powered features on the Claude API** — resume feedback, skill extraction, job-fit recommendations, interview prep, and cover-letter generation — integrated directly into candidate and employer workflows, replacing a manual shortlisting process.
- Shipped the platform to production**; now in **active daily use by the company**, holding a **5.0-star rating across 19 Google reviews**, while working directly with **3 concurrent client stakeholders** to translate requirements into shipped features via Agile iteration in Trello.

DVertex Info System (Safaimitra)

Nov 2024 – Mar 2025

Software Development Intern

Remote

- Resolved **15+ bugs** in Safaimitra's live, event-driven push-notification system (**AWS SNS/Lambda**) and other core modules of the e-waste management app, working within an Agile/Scrum team.
- Ran **UAT/QA** across releases to catch regressions before they reached production, keeping releases stable for live users.

Bell (Best Buy Express)

Oct 2024 – Nov 2025

Technology & Sales Associate (Part-time)

Kelowna, BC

- Ranked 2nd in BC** for phone plan sales volume; consistently hit **100%+ of monthly targets (2x quota in June)** while maintaining KYC compliance in a regulated, high-volume retail environment.

Projects

RAG Evaluation Pipeline

2026

Python, sentence-transformers, ChromaDB, Claude API, Streamlit

GitHub

- Architected** a full **Retrieval-Augmented Generation** pipeline with chunking, embedding generation, vector storage (ChromaDB), cross-encoder reranking, and cited answer generation, built over **4,500+** chunks from 51 Wikipedia articles, with pytest unit tests and GitHub Actions CI for regression safety, plus a Streamlit demo UI.
- Built a **100-question, human-reviewed** evaluation harness rather than relying on a single synthetic metric; the reranking pipeline matched a **BM25 baseline** I also built, both hitting **97% Recall@5** on the initial eval set.
- Pressure-tested** that result for overfitting by building a second, paraphrased eval set: BM25 recall collapsed to 19% while the embedding pipeline held at **54% (2.8x more robust)**, exposing a hidden keyword-matching bias in the original benchmark that a single eval set would have missed.
- Layered in **LLM-as-judge** grading for answer-level correctness (**83%/93% accuracy** across two grading passes), catching failure modes that retrieval-recall metrics alone couldn't surface.

Rate Limiter Lab

2026

Python, FastAPI, Redis (Lua scripting), asyncio, pytest

GitHub

- Implemented **3 Redis-backed rate-limiting algorithms** (Fixed Window, Sliding Window, Token Bucket), each in a naive and an atomic (Lua-script) version, exposed via FastAPI endpoints.
- Identified and reproduced a genuine race condition** under concurrent load: naive implementations over-admitted requests on **20/20 test bursts**, letting up to 40 requests through a limit of 10.
- Fixed** the race condition with atomic Lua scripting and proved it with a dedicated regression test, driving failures from 20/20 down to **0/20** under the same concurrent load.
- Benchmarked** the fixed algorithms under concurrent HTTP load (throughput, p50/p95/p99 latency), with CI running tests against a live Redis service container to keep results reproducible.

Skills

Languages: Python, JavaScript, TypeScript, Java, PHP, HTML/CSS, SQL

Backend & APIs: FastAPI, Node.js (Next.js API routes), REST API design, OAuth 2.0, PostgreSQL, Supabase, SQLAlchemy, Redis, asyncio

AI/LLM: Claude API, LLM-integrated product features, RAG pipelines, embeddings, vector databases (ChromaDB), prompt engineering

Frontend: React, Next.js

DevOps & Tooling: Docker, Docker Compose, Git, GitHub Actions (CI/CD), pytest, JUnit, Alembic

Cloud: AWS (SNS, Lambda), Render, Vercel

Practices: Agile/Scrum, Jira, Confluence, Trello